

AI PROJECT

TEAM MEMBERS:

SOURISH DUTTA (24BCT0282), PRATHAMESH CHALAK(24BCT0286)

ABSTRACT:

TITLE: Explainable Autonomous Game Agents Using Deep Reinforcement Learning and Retrieval-Augmented Generation (RAG)

As AI gets better at playing games and making fast decisions, understanding *why* an agent acts the way it does has become just as important as the action itself. Deep learning models are great at adapting under pressure, but when an agent suddenly changes tactics—like running away from a fight or choosing a weird route—it's usually impossible to tell what made it make that call.

The problem is that traditional AI models are total black boxes. You can't just feed raw game numbers and thousands of data points into a standard chat AI like ChatGPT and ask for an explanation. All those raw numbers instantly overwhelm the model, causing it to hallucinate and make up believable-sounding reasons that are completely wrong.

To fix this, our project combines smart game AI with a quick, factual explanation system. We built a custom 2D arena game using Pygame and Gymnasium, where an AI agent learns to fight, dodge, and navigate using Deep Reinforcement Learning (specifically, PPO). As the agent plays, the game constantly logs its exact status—like its health, position, and active rewards—into a ChromaDB vector database. If you pause and ask why the agent made a specific move, a Retrieval-Augmented Generation (RAG) system grabs that exact frame of data and uses a local language model to explain the move in simple English.

We're testing our system by tracking how well the AI learns in TensorBoard, checking how fast the database can pull up log files, and making sure the explanations match what actually happened in the game without making things up.

Overall, this project shows a practical way to peek inside an AI's head without breaking its workflow. While it's great for tuning game bots and balancing game mechanics, this kind of transparent AI also has huge potential for military training simulations and making real-world autonomous systems much safer and easier to trust.